10 Two satellites P and Q orbit the Earth due to the gravitational field of the Earth. P and Q are at distances R and $3R$ from the Earth's surface respectively, where R is the radius of the earth.

The speed of Q is v.

What is the speed of P?

A $\sqrt{\frac{1}{3}}\,v$ **B** $\sqrt{\frac{1}{2}}\,v$ **C** $\sqrt{2}\,v$ **D** $\sqrt{3}\,v$